



United International University

Department of Computer Science and Engineering

Introduction to Programming Language II (Java)

Topic: Variables, I/O, & Strings

Duration: 2.5 hours

Lab Sheet: Introduction to Java Programming

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1. Objectives

By the end of this lab, students will be able to:

- Understand Java program structure and the `main` method.
- Declare and initialize variables of primitive and reference types (`int`, `double`, `String`).
- Read user input using the `Scanner` class and print formatted console outputs.
- Implement string concatenation and basic arithmetic mutations to build interactive systems.

2. Introduction to Java Basics

Java is a high-level, object-oriented programming language. Every execution sequence begins with a container class housing a predefined structural entryway: the `main` method.

Key Components:

- **Variables:** Logical tokens holding primitive or referenced values.
- **Data Types:** Structural parameters defining byte ranges (`int`, `double`) or textual references (`String`).
- **I/O Streams:** Processing pipelines utilizing `Scanner` for runtime input and `System.out` for terminal views.

3. Common Data Types

- **int:** Stores whole numerical blocks without fraction steps (e.g., 42).
- **double:** Stores high-precision structural floating decimal points (e.g., 3.75).
- **String:** Encapsulates an ordered collection of character data literals (e.g., "Hello, World!").

4. Example 1: Variable Declaration and Initialization

VariableDeclaration.java

```
public class VariableDeclaration {
    public static void main(String[] args) {
        // Variable declaration and initialization
        String studentName = "John Doe";
        int studentId = 12345;
        double gpa = 3.75;

        // Display all variables
        System.out.println("Student Name: " + studentName);
        System.out.println("Student ID: " + studentId);
        System.out.println("GPA: " + gpa);
    }
}
```

5. Example 2: Printing Output

PrintOutput.java

```
public class PrintOutput {  
    public static void main(String[] args) {  
        String message = "Hello, World!";  
        int number = 42;  
  
        System.out.println("Simple message: " + message);  
        System.out.println("Number: " + number);  
        System.out.println("The answer is: " + number);  
    }  
}
```

6. Example 3: String Concatenation

StringConcatenation.java

```
public class StringConcatenation {  
    public static void main(String[] args) {  
        String firstName = "Alice";  
        String lastName = "Smith";  
  
        String fullName = firstName + " " + lastName;  
        System.out.println("Full Name: " + fullName);  
  
        int age = 20;  
        System.out.println("Name: " + fullName + ", Age: " + age);  
    }  
}
```

7. Example 4: Reading Input with Scanner

ScannerInput.java

```
import java.util.Scanner;  
  
public class ScannerInput {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter your name: ");  
        String name = sc.nextLine();  
        System.out.println("You entered: " + name);  
  
        sc.close();  
    }  
}
```

8. Example 5: Integer Data Type

IntegerDataType.java

```
import java.util.Scanner;

public class IntegerDataType {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        int number = sc.nextInt();

        System.out.println("Integer value: " + number);
        System.out.println("Double the value: " + (number * 2));

        sc.close();
    }
}
```

9. Example 6: Double Data Type

DoubleDataType.java

```
import java.util.Scanner;

public class DoubleDataType {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a decimal number: ");
        double decimal = sc.nextDouble();

        System.out.println("Double value: " + decimal);
        System.out.println("Rounded value: " + Math.round(decimal));

        sc.close();
    }
}
```

10. Example 7: String Data Type

StringDataType.java

```
import java.util.Scanner;

public class StringDataType {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a text: ");
        String text = sc.nextLine();

        System.out.println("String value: " + text);
        System.out.println("Length: " + text.length());

        sc.close();
    }
}
```

11. Tasks & Solutions

Task 1: Variable Declaration (10 min)

- Create a class `StudentInfo` that declares variables for name (`String`), roll number (`int`), and CGPA (`double`).
- Initialize them with sample values and print them to the terminal.

Solution:

StudentInfo.java

```
public class StudentInfo {  
    public static void main(String[] args) {  
        String name = "Alice Johnson";  
        int roll = 202301;  
        double cgpa = 3.85;  
  
        System.out.println("Student Name: " + name);  
        System.out.println("Roll Number: " + roll);  
        System.out.println("CGPA: " + cgpa);  
    }  
}
```

Task 2: User Input Program (15 min)

- Modify your runtime configurations to process clean structural inputs with descriptive execution strings.
- Parse user strings, ID fields, and decimal configurations sequentially within an execution scope.

Solution:

StudentInput.java

```
import java.util.Scanner;

public class StudentInput {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your name: ");
        String name = sc.nextLine();
        System.out.print("Enter your ID: ");
        int id = sc.nextInt();
        System.out.print("Enter your CGPA: ");
        double cgpa = sc.nextDouble();

        System.out.println("Name : " + name);
        System.out.println("ID : " + id);
        System.out.println("CGPA : " + cgpa);

        sc.close();
    }
}
```

Task 3: Interactive Calculator (20 min)

- Take two structural values alongside a targeted control operator argument via input sequences.
- Implement basic arithmetic procedures through your preferred numeric type boundaries (`int` or `double`).
- Structure output rendering patterns cleanly to mirror logical operations.

Task 4: Comprehensive Portfolio Generation (25 min) Synthesize the foundational principles reviewed during today's session:

- Capture runtime targets representing multi-part personal identities (`firstName`, `lastName`), absolute milestones (`age`), and performance configurations (`gpa`).
- Output a formatted student profile summary to the console window.
- Compute intermediate programmatic changes to evaluate updated age values alongside rounded performance records.

Keep Coding!

Mastering these fundamentals is the foundation for writing robust Java applications.